3.7.4 Capacitance (A-level only)

3.7.4.1 Capacitance (A-level only)

Content	Opportunities for skills development
Definition of capacitance: $C = \frac{Q}{V}$	

3.7.4.2 Parallel plate capacitor (A-level only)

Content	Opportunities for skills development
Dielectric action in a capacitor $C = \frac{A\varepsilon_0\varepsilon_r}{d}$ Relative permittivity and dielectric constant. Students should be able to describe the action of a simple polar molecule that rotates in the presence of an electric field.	**PS 1.2, 2.2, 4.3 / AT f, g** Determine the relative permittivity of a dielectric using a parallel-plate capacitor. Investigate the relationship between C and the dimensions of a parallel-plate capacitor eg using a capacitance meter.

3.7.4.3 Energy stored by a capacitor (A-level only)

Content	Opportunities for skills development
Interpretation of the area under a graph of charge against pd. $E = \frac{1}{2}QV = \frac{1}{2}CV^2 = \frac{1}{2}\frac{Q^2}{C}$	

3.7.4.4 Capacitor charge and discharge (A-level only)

Content	Opportunities for skills development
Graphical representation of charging and discharging of capacitors through resistors. Corresponding graphs for Q, V and I against time for charging and discharging. Interpretation of gradients and areas under graphs where appropriate. Time constant RC. Calculation of time constants including their determination from graphical data. Time to halve, $T_{½} = 0.69RC$ Quantitative treatment of capacitor discharge, $Q = Q_0 e^{-\frac{t}{RC}}$ Use of the corresponding equations for V and I. Quantitative treatment of capacitor charge, $Q = Q_0\left(1 - e^{-\frac{t}{RC}}\right)$	
Required practical 9: Investigation of the charge and discharge of capacitors. Analysis techniques should include log-linear plotting leading to a determination of the time constant, RC	**MS 3.8, 3.10, 3.11 / PS 2.2, 2.3 / AT f, k**